Yumi Zhao

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EDUCATION

University of California, Berkeley **Dec. 2027**

B.S. in BioEngineering (Device / Systems emphasis) | GPA: 3.95/4.0

HONORS:

Regents' and Chancellor's Scholar and Yardi Scholar: Highest undergraduate honor awarded to top ~1–2% of students.

Berkeley Engineering Design Scholar: Selected for a highly competitive summer research fellowship by Jacobs Institute.

RESEARCH EXPERIENCE

UC Berkeley, Ornithopterus Lab | Berkeley, CA **Aug. 2025 - Present**

SMA-Based Actuator Lab Experience *PI: Professor Víctor M. Ortega Jiménez*

- Designed and fabricated a custom mechanical wind-tunnel release fixture using a pulley-and-counterweight architecture, eliminating airflow interference to capture instantaneous takeoff acceleration.
- Participated in physical prototype assembly and mechanical iteration for bio-inspired robotic mechanisms; integrated SMA (Shape Memory Alloy) actuators and optimized wiring runs to reduce structural interference.
- Implemented embedded firmware on ATtiny85/Seeeduino to execute actuation profiles and stream real-time sensor telemetry over serial UART to MATLAB.
- Assembled and debugged custom driver circuits on breadboards and deadbug soldered PCBs, optimizing power conditioning and signal routing for SMA thermal cycling.
- Standardized hardware setup and testing protocols by authoring comprehensive lab assembly and wiring guides, enabling repeatable physical testing for ongoing research.

PROJECTS

Closed-Loop Microfluidic Sensing & Automation Software | Berkeley Engineering Design Scholar | *PI: Professor Hayden Taylor* **July. 2026**

- **Architected and developed a modular, single-threaded Python/PyQt software platform** with custom API interfaces for real-time microfluidic velocity monitoring and diffusion-coefficient estimation.
- **Designed an automated ROI tracking pipeline** validated across 42 test trials, integrating an adaptive frame-rate control algorithm to bypass hardware sampling bottlenecks.
- **Implemented intensity-based physics algorithms** to estimate diffusion metrics, achieving a mean percentage error of ~10% (<20% max error threshold) against theoretical physical models.
- **Engineered a fault-tolerant DAQ pipeline** with periodic 5-minute CSV checkpointing and custom memory management, enabling 10+ hours of continuous, leak-free operation.

Closed-Loop Environmental Control & DAQ Platform | *PI: Professor Zahra Noroozi* **November. 2024**

- Built an embedded environmental control system using Arduino, thermistors, photoresistors, and micro-servos to execute closed-loop temperature and lighting regulation in real time.
- Designed a MATLAB App Designer GUI for high-frequency data logging, dynamic serial visualization, customizable setpoints, and automated Day/Night mode execution.

Active Bandpass Filter & Op-Amp Signal Conditioning System | *EE 16B Course Project* **August.2026**

- Designed and charactered an Op-Amp active bandpass filter circuit to attenuate high-frequency noise and out-of-band interference, boosting target frequency gain for robust signal acquisition.
- Performed frequency response analysis (Bode Plots) and hardware debugging on breadboard, validating theoretical transfer functions with real-world sensor input signals.

SKILLS & RELEVANT COURSEWORK

- **Languages & Frameworks:** Python (PyQt), MATLAB (App Designer/Simulink), Java, SQL, Git/GitHub.
- **Hardware & Embedded:** Arduino (Uno/ATtiny/Seeeduino), Embedded System Prototyping, Serial Communication, Sensor Integration, DAQ, Precision Soldering (Deadbug), Op-Amp Circuits.
- **Prototyping & CAD:** 3D Parametric CAD (Rhino), Spatial Modeling, Microfluidics Assembly, Precision Soldering (Deadbug/PCB), Component Diagnostics.
- **Control, Signals & Robotics:** Forward/Inverse Kinematics, Manipulator Dynamics, Robotic Vision, LTI Systems, Bode Plots, Feedback Control Stability, Statistical Data Analysis.